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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 3: Concept of Sustainability

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Contents

- What makes a development to be unsustainable?
- Concept of Sustainable Development
- Goals for Sustainable Development
- Sustainable Transport: Goals, Objectives & Examples
- Conclusion



Source: (N. Mohan Das Gandhi et. al, 2006)

Present Course of Development: Sustainable or Unsustainable ?

- Human's life on Earth has been facing sustainability related challenges, especially since the beginning of the Industrial Revolution. Though, industrialization contributed a lot in terms of population growth, employment generation and wealth creation.
- Pollution has been an essential by-product of the present fossil fuels based economic system.

Some thought leaders argue that the development trajectories through which Humans have evolved are at the cost of their own future generations, thus it is Unsustainable in the real sense.



Source: (N. Mohan Das Gandhi et. al, 2006)

Factors contributing to Unsustainable Development

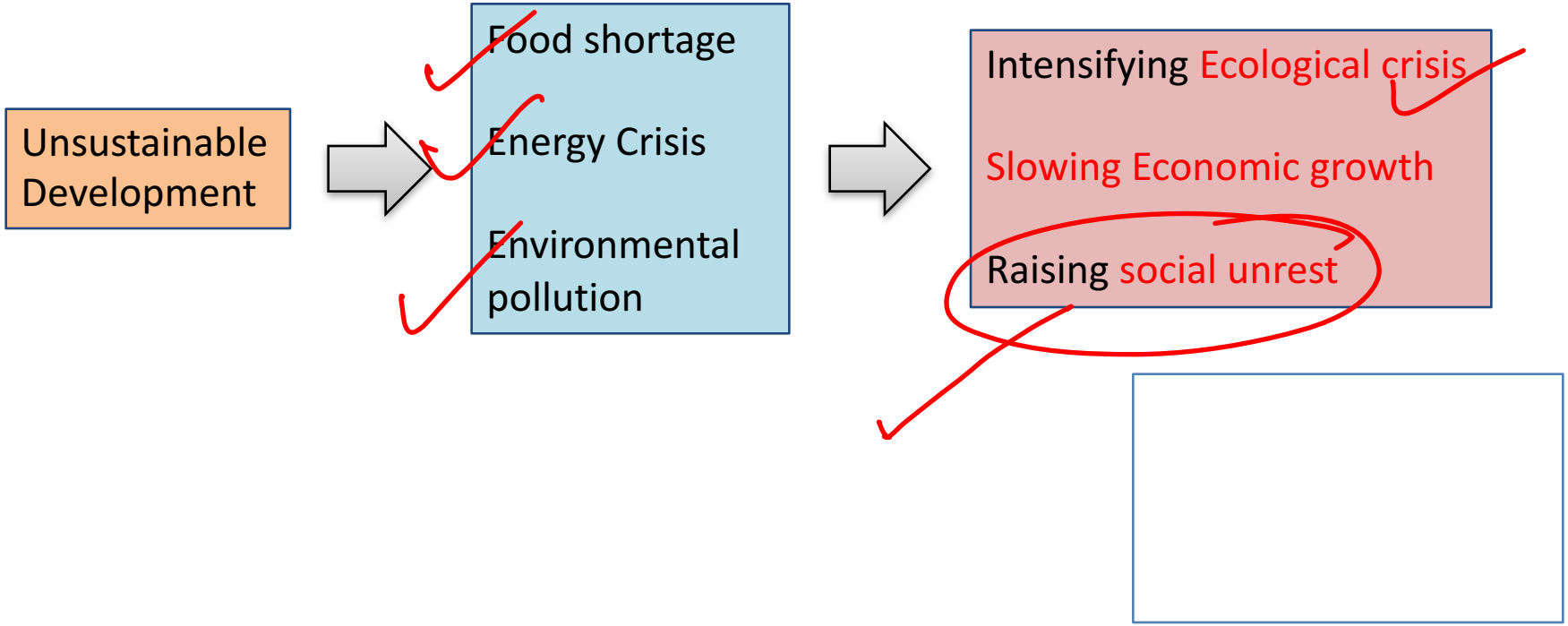
- Polluting industrialization
- Increasing economy and per capita income with huge gap between rich and poor
- Unchecked Population Growth
- Continuous depletion of Non Renewable resources
- Climate Change



Negative Externalities of an Unsustainable Development

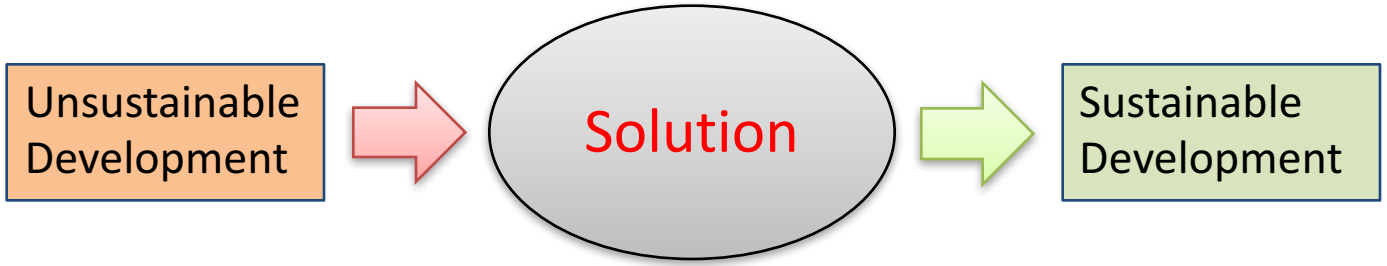
- Air pollution
- Noise pollution
- Accidents
- Global warming
- Depletion of Energy Resources
- Congestion
- Social disruption
- Consumption of land
- Loss of habitat
- Hazardous materials
- Vibration
- Visual intrusion
- Waste disposal
- Water pollution

What are the Impacts ?



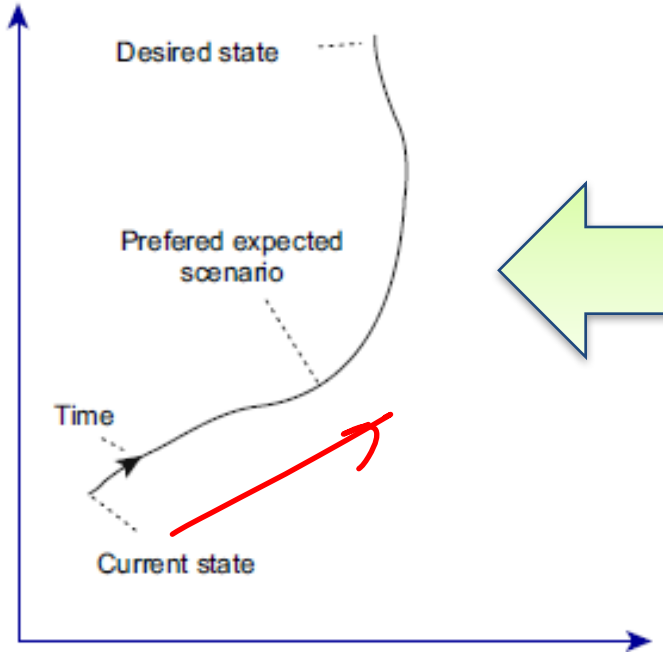
Source: (Longyu Shi et. al, 2019)

What is a Sustainable Development ?



One of the very popular definitions of Sustainable Development illustrates; “The development that meets the needs of the present without compromising the ability of future generations to meet their own needs” *(Brundtland-report, 1987).*

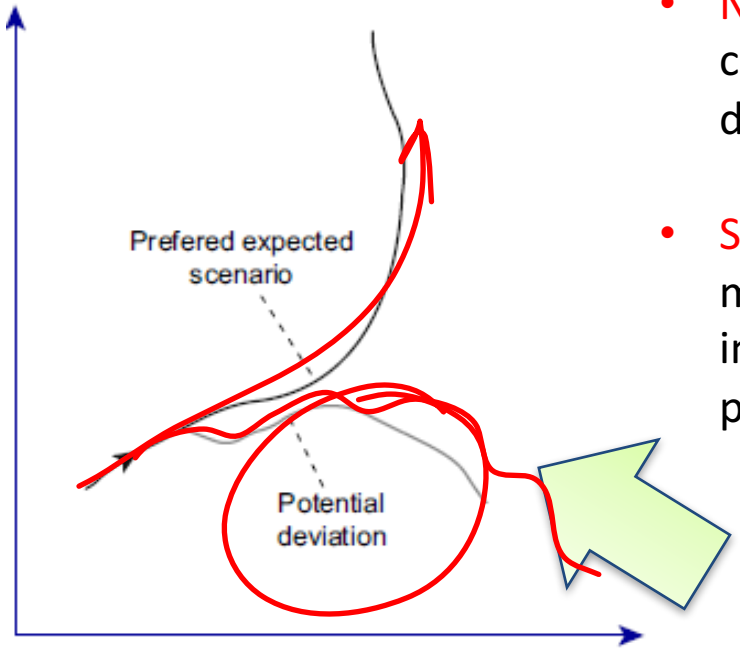
The Concept of Development



Development as a preferred expected scenario of change in a set of variables over time, from a current to a desired state, including purposeful activity

Source: (Per Becker, 2014)

Negative externalities and Development

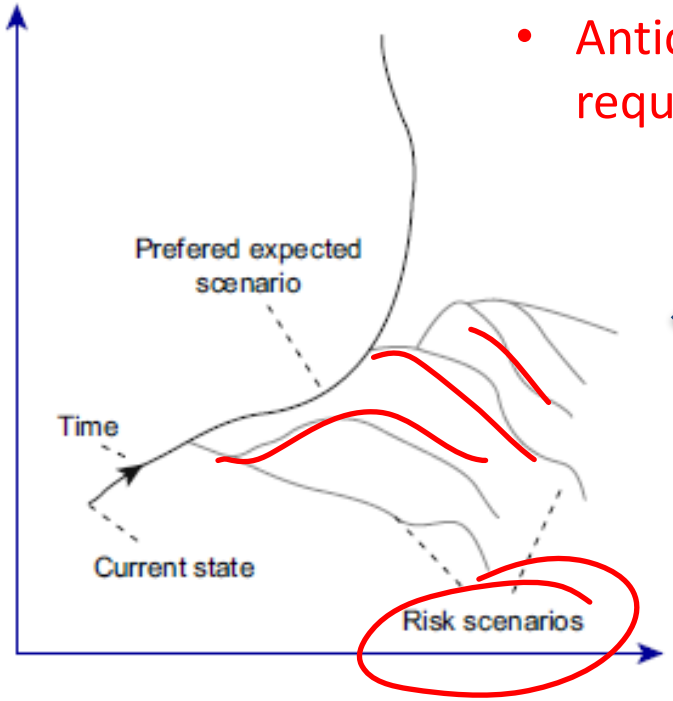


- Negative events and their underlying processes may cause deviations from our preferred expected development scenario, limiting our sustainability.
- Sustainable development is the development that can be maintained over time and be safeguarded from the impact of negative events and their underlying processes.

Potential deviation from the preferred expected Development scenario

Source: (Per Becker, 2014)

The Concept of Sustainable Development

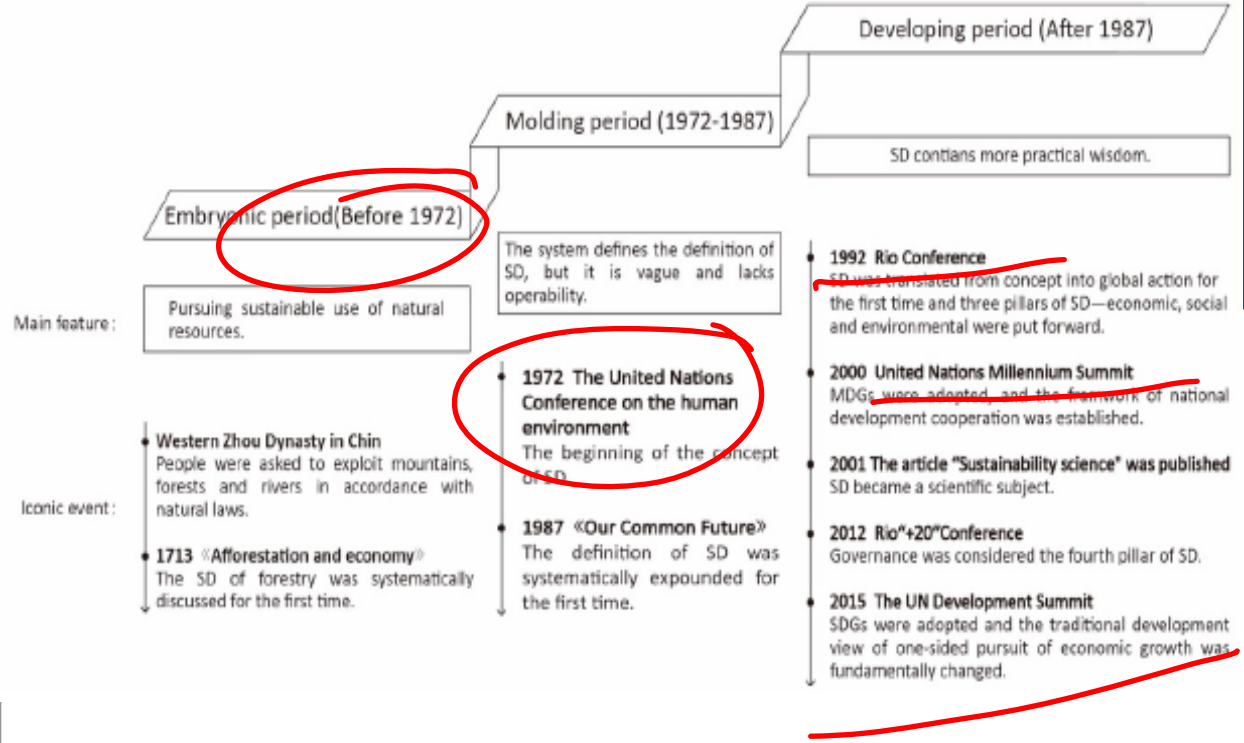


- Anticipating potential deviations, or risk scenarios, is a requisite for sustainable development

The Preferred expected scenario as a Sustainable development trajectory of the human-environment system

Source: (Per Becker, 2014)

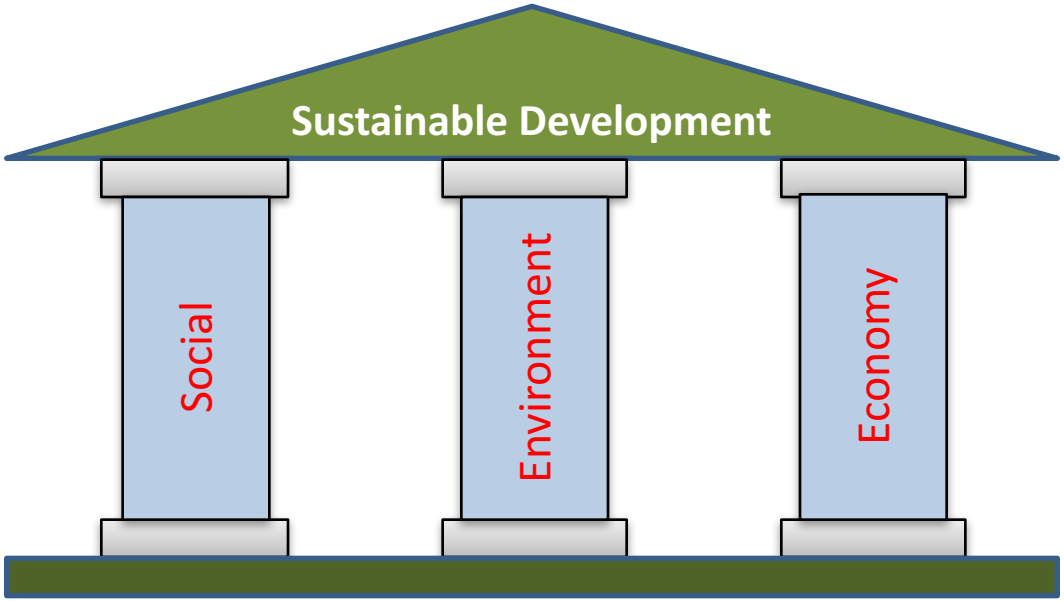
Evolution of the term “Sustainable Development”



- The Embryonic Period (Before 1972)
- The Molding Period (1972-1987)
- The Developing Period (After 1987)

Source: (Longyu Shi et. al, 2019)

The Three Pillars of Sustainability

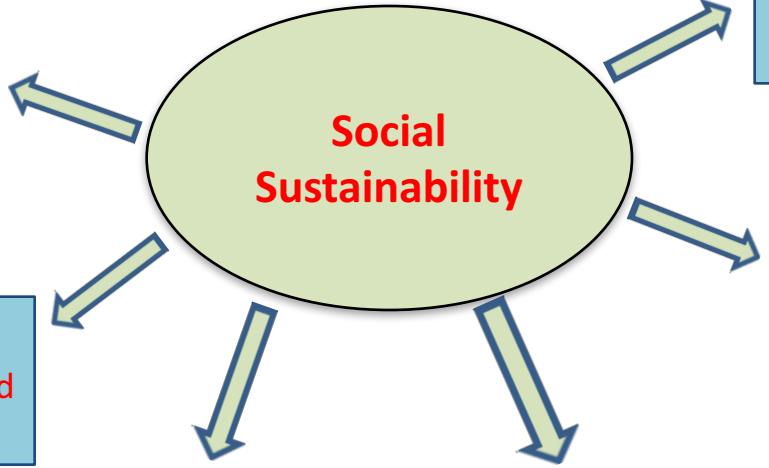


Sustainability can be of three types:

- Social sustainability
- Environmental sustainability
- Economic sustainability

Source: (IUCN, The World Conservation Union, 2006)

Social Sustainability: Criteria



Environmental Justice
Protect health of communities over-burdened by pollution by empowering them to take action to improve their health and environment

Sustainable Communities
Promote the development, planning, building, or modification of communities to promote sustainable living

Resource Security
Protect, maintain, and restore access to basic resources
Ex. food, land, and energy

Human health
Protect, sustain, and improve human health

Participation
Use open and transparent processes that engage relevant stakeholders

Education
Enhance the education about sustainability

Source: (EPA, https://www.epa.gov/sites/production/files/2015-05/documents/sustainability_primer_v7.pdf)

Environmental Sustainability: Criteria



Ecosystem Services

Protect, sustain, and restore the health of critical natural habitats and ecosystems

Resource Integrity

Reduce adverse effects by reducing waste generation, increased recycling, and ensuring proper waste management; restore resources by mitigating and cleaning up accidental or intentional releases

Green Engineering and Chemistry

Develop chemical products and processes to reduce/prevent chemical hazards, reuse or recycle chemicals, treat chemicals to render them less hazardous, dispose of chemicals properly.

Stressors

Reduce effects by stressors to the ecosystem

Ex. Pollutants, GHG emissions, genetically modified organisms

Air Quality

Attain and maintain air-quality standards and reduce the risk from toxic air pollutants

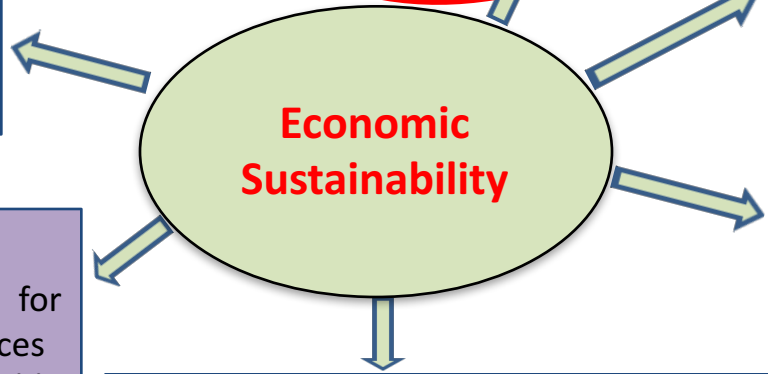
Water Quality

Reduce exposure to contaminants in water bodies

Source: EPA, (https://www.epa.gov/sites/production/files/2015-05/documents/sustainability_primer_v7.pdf)

Economic Sustainability:

Criteria



Prices

Promote a cost structure that accounts for externalities to production

Cost

Positively impact costs of processes, services, and products

Jobs

Create or maintain job opportunities
Ex. Create green jobs

Natural Resource Accounting

Incorporate natural capital depreciation in accounting indices and ecosystem services in cost benefit analysis.
Ex. Green net national product

Incentives

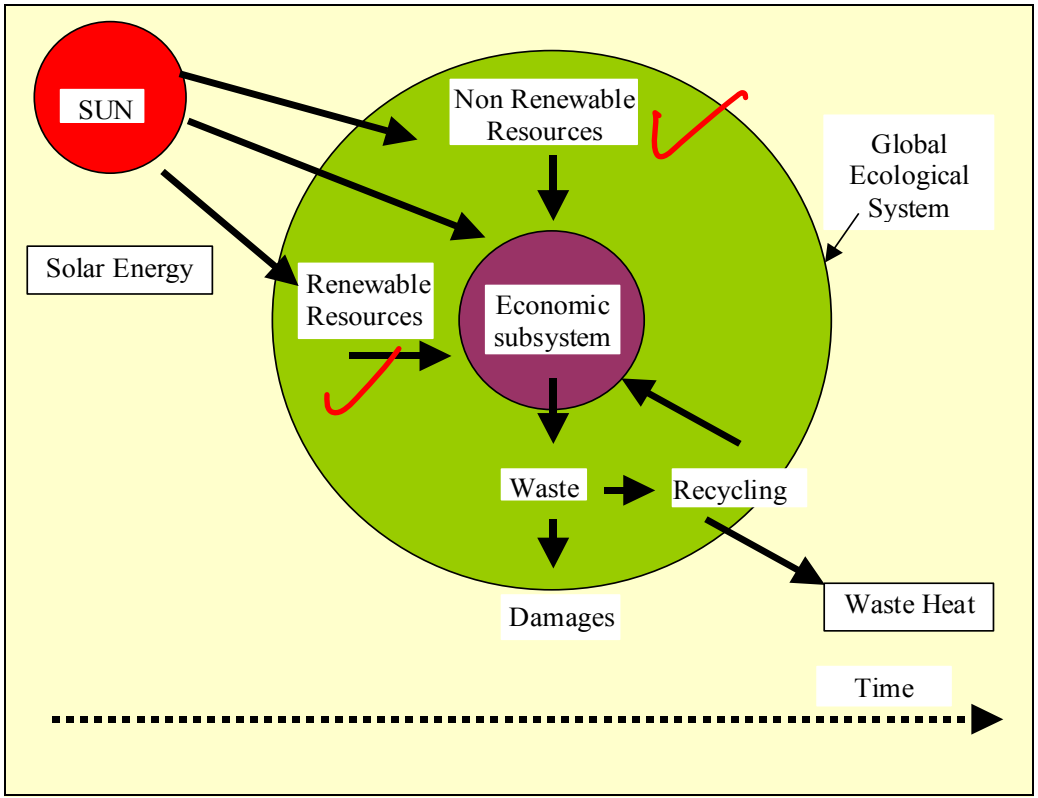
Generate incentives for encouraging sustainable practices
Ex. Encouraging sustainable logging practices

Supply and Demand

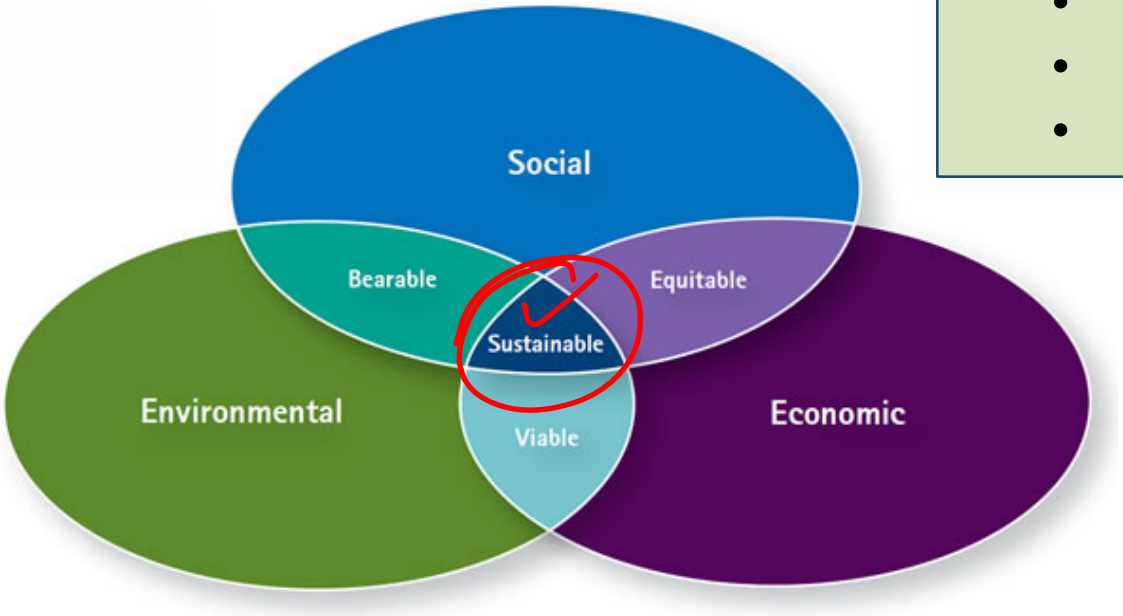
Promote price or quantity changes that alter economic growth, environmental health and social prosperity.
Ex. Increasing supply of green energy sources to reduce need for fossil fuels

Source: EPA, (https://www.epa.gov/sites/production/files/2015-05/documents/sustainability_primer_v7.pdf)

Interaction of Economy on Ecological System



Principles of Sustainability

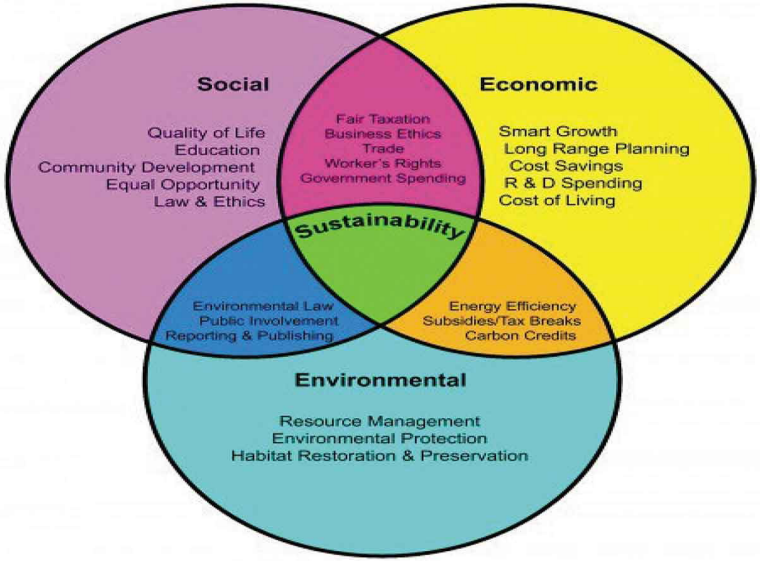


Sustainable Development should be:

- Environmentally bearable
- Socially equitable
- Economically viable

Basic Principles of Sustainability

- Minimizing use of non-renewable resources
- Minimizing impacts on natural environment
- Protecting biodiversity
- Using renewable resources in a Sustainable manner



Source: (Justice Mensah, 2019)

Goals for Sustainable development (SDGs)



Source: (UNDESA, https://www.un.org/development/desa/disabilities/envision2030.html)

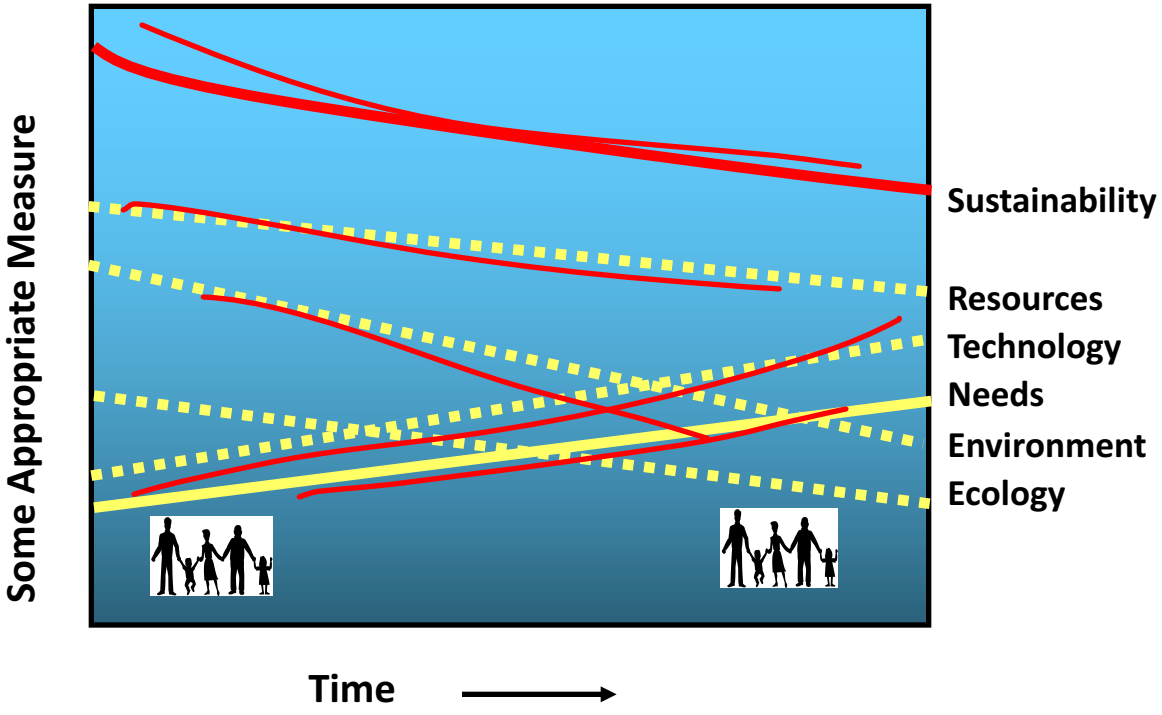
Sustainable Transport



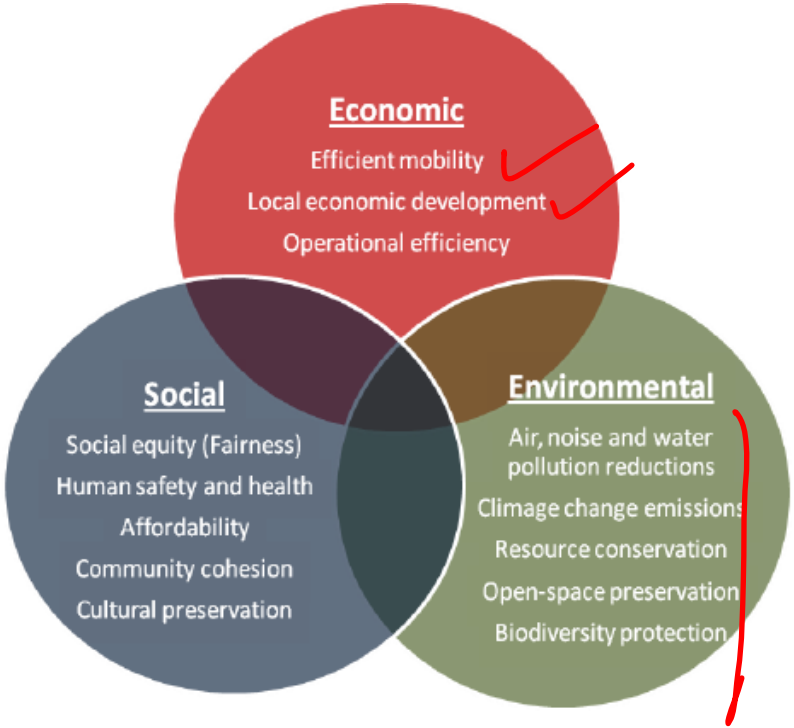
Sustainable transport is the provision of services and infrastructure for the mobility of people and goods, advancing economic and social development to benefit today's and future generations in a manner that is safe, affordable, accessible, efficient, and resilient, while minimizing carbon and other emissions and environmental impacts.

Source: (UNDESA, 2014)

Defining Sustainable Transportation



Sustainable Transport Goals











Source: (UNDESA, 2014)

Sustainable Transportation Goals

Dimension	Goals
Social	Reduce Congestion Enhance Safety
Economic	Expand Economic Opportunity Increase Value of Transportation Assets
Environmental	Improve Air Quality



Sustainable Transportation Goals

Sustainable Transport is:

- Safe
  - Good health and wellbeing (SDG 3)
- Affordable
- Accessible
  - Zero Hunger (SDG 2)
- Efficient
  - Affordable and Clean Energy (SDG 7)
  - Responsible consumption and production (SDG 12)









Source: (UNDESA, 2014)

Sustainable Transportation Goals

Sustainable Transport is:

- Resilient
  - Industry Innovation and Infrastructure (SDG 9)
- Minimizing Carbon and other emissions and Environmental impacts
  - Climate Action (SDG 13)
  - Sustainable cities and Communities (SDG 11)







Source: (UNDESA, 2014)

Sustainable Transport: Objectives

The "5 Is and 5 Cs" approach

The 5 I's of sustainable transport *areas of action:*	The 5 C's of sustainable transport *to meet the needs of:*
Infrastructure enabling services	**Client** Meeting expectations with quality services
Innovation for the future	**Centres** Passenger and logistic hubs
Integration of modes	**Corridors** Connecting the hubs
Intelligence Capacity building & training	**Congestion** Low pollution improved efficiency
Investment to create a virtuous cycle	**Complementarity** Optimized multimodal transport systems

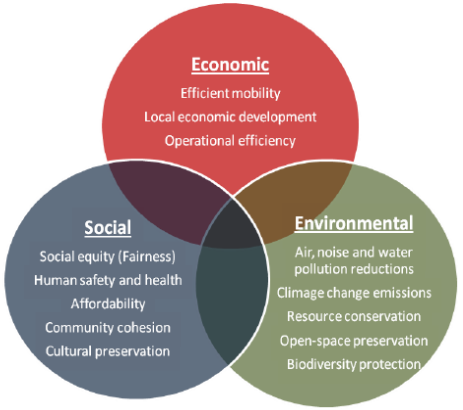


Source: (UNDESA, 2014)

Sustainable Transport: Goals, Objectives & Indicators

Economic Sustainability

Sustainability Goals	Objectives	Performance Indicators
I. Economic		
Economic productivity	Transport system efficiency. Transport system integration. Maximize accessibility. Efficient pricing and incentives.	• Per capita GDP • Portion of budgets devoted to transport. • Per capita congestion delay. • Efficient pricing (road, parking, insurance, fuel, etc). • Efficient prioritization of facilities
Economic development	Economic and business development	• Access to education and employment opportunities. • Support for local industries.
Energy efficiency	Minimize energy costs, particularly petroleum imports.	• Per capita transport energy consumption • Per capita use of imported fuels.
Affordability	All residents can afford access to basic (essential) services and activities.	• Availability and quality of affordable modes (walking, cycling, ridesharing and public transport). • Portion of low-income households that spend more than 20% of budgets on transport.
Efficient transport operations	Efficient operations and asset management maximizes cost efficiency.	• Performance audit results. • Service delivery unit costs compared with peers. • Service quality.

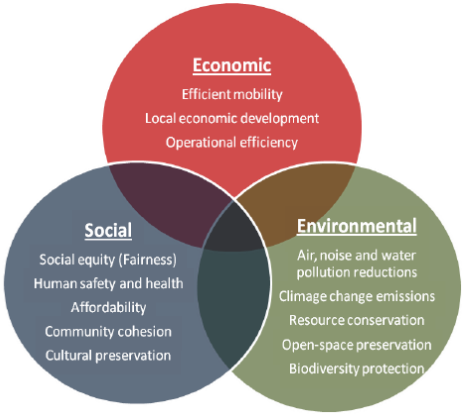


Source: (Todd Litman, 2021)

Sustainable Transport: Goals, Objectives & Indicators

Social Sustainability

Sustainability Goals	Objectives	Performance Indicators
II. Social		
Equity / fairness	Transport system accommodates all users, including those with disabilities, low incomes, and other constraints.	• Transport system diversity. • Portion of destinations accessible by people with disabilities and low incomes.
Safety, security and health	Minimize risk of crashes and assaults, and support physical fitness.	• Per capita traffic casualty (injury and death) rates. • Traveler assault (crime) rates. • Human exposure to harmful pollutants. • Portion of travel by walking and cycling.
Community development	Helps create inclusive and attractive communities.	• Land use mix. • Walkability and bikability • Quality of road and street environments.
Cultural heritage preservation	Respect and protect cultural heritage. Support cultural activities.	• Preservation of cultural resources and traditions. • Responsiveness to traditional communities.

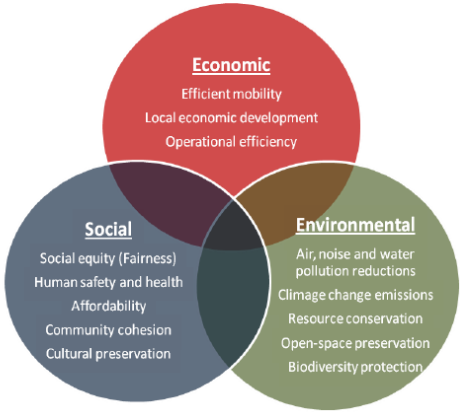


Source: (Todd Litman, 2021)

Sustainable Transport: Goals, Objectives & Indicators

Environmental Sustainability

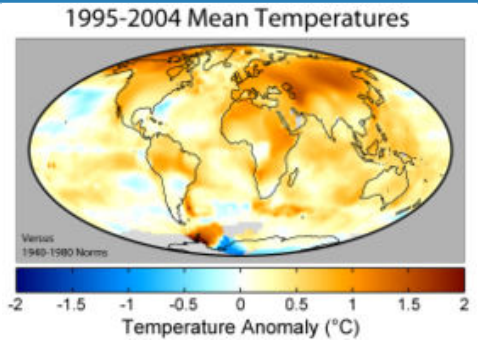
Sustainability Goals	Objectives	Performance Indicators
III. Environmental		
Climate stability	Reduce global warming emissions Mitigate climate change impacts	• Per capita emissions of greenhouse gases (CO_2, CFCs, CH_4, etc.).
Prevent air pollution	Reduce air pollution emissions Reduce harmful pollutant exposure	• Per capita emissions (PM, VOCs, NOx, CO, etc.). • Air quality standards and management plans.
Minimize noise	Minimize traffic noise exposure	• Traffic noise levels
Protect water quality & hydrologic functions	Minimize water pollution. Minimize impervious surface area.	• Per capita fuel consumption. • Management of used oil, leaks and stormwater. • Per capita impervious surface area.
Openspace and biodiversity protection	Minimize transport facility land use. Encourage compact development. Preserve high quality habitat.	• Per capita land devoted to transport facilities. • Support for smart growth development. • Policies to protect high value farmlands and habitat.
IV. Good Governance and Planning		
Integrated, comprehensive and inclusive planning	Clearly defined planning process. Integrated and comprehensive analysis. Strong citizen engagement. Lease-cost planning.	• Clearly defined goals, objectives and indicators. • Availability of planning information and documents. • Portion of population engaged in planning decisions. • Range of objectives, impacts and options considered. • Efficient and equitable funding allocation

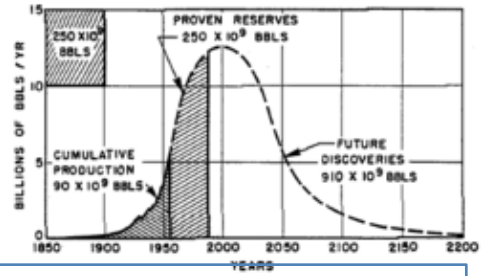


Source: (https://www.vtpi.org/wellmeas.pdf)

What is Changing? Why this emphasis now?

- Growing awareness of problems or issues
- Increasing intolerance with use of “free goods or services”
- Mistrust of global organizations
- Information age has increased power of third parties.
- Expectation that something be done





Sustainable Development: A Few Examples

Octopus Smart Card, Hong Kong

- 93% of transport trips are made by public transport (48%) or walking (45%).

- The Octopus Smart Card is a contactless smart card payment system that started as a public transport fare collection system in 1997.

- Used for Access control for residential and commercial buildings, integrated public transportation modes, payment and settlement purposes, schools etc.



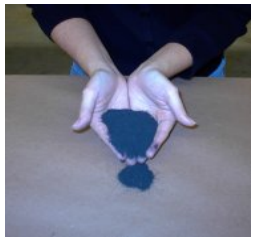


Source: (UNDESA, 2014)

Sustainable Development: A Few Examples

Reduce, Recycle, Reuse

- Grey County, Ontario.
  - Issue – Thousands of used car tires.
  - Solution – Build a facility to recycle the tires and offer the rubber for utilization in the production of asphalt pavements within Grey county.


Source: (AMEC Case study, 2007)

Sustainable Development: A Few Examples

The Last Mile Challenge



- Inclusive Transport for Low-Density Rural Areas, Chille
  - "Leave no one behind" – Included in the 2030 Agenda for Sustainable Development
  - To connect the rural areas, disconnected from the major roads, rail and other public transport modes.
  - Ex. Inclusive waterborne transport in extremely low-populated areas in the southern parts of Chile, Free bus transport for handicapped children from remote areas throughout the country.

Source: (UNDESA, 2014)

Sustainable Development: A Few Examples

Dedicated Freight Corridor, India



- One of the largest transport infrastructure projects implemented in India.
- A major transition in the freight transport sector by increasing the relative share of rail as an energy efficient, environment friendly and less carbon-intensive mode of transport.
- It is expected that DFC will save more than 450 million tons of CO_2 in first 30 years of operation.

Source: (UNDESA, 2014)

Sustainable Development: A Few Examples

Eastern Canada Road Weather Information Systems (RWIS)

- Economically beneficial through the reduction in the use of road salt.
- Increases driver safety due to improved salt management
- Lessens the environmental impact on road side vegetation and wildlife









Source: (AMEC Case study, 2007)

Sustainable Development: A Few Examples

Bus Rapid Transit System, Johannesburg



- The designation of Johannesburg as one of the hosts of the 2010 FIFA World Cup in South Africa provided the impetus to improve public transport in the city.
- The first full Bus Rapid Transit (BRT) system in Africa.
- The total cost: US$749 million
- The economic returns estimated (2026): US$900 million

Source: (UNDESA, 2014)

Sustainable Development: A Few Examples



Transparent Policy Progress in Copenhagen

- Aims to become the "eco-metropolis" of the world and the world's best city for cyclists, and to be CO_2 -neutral by 2025.
- Today, 96% of Copenhagen are able to walk to a large green or blue recreational area in less than 15 minutes.
- Over 50% of people uses cycle to their workplace or educational institutions.
- Noise-reducing asphalt, traffic planning and speed reduction have reduced traffic noise in general in the city.

Source: (UNDESA, 2014)

Sustainable Aggregate Recycling: Europe vs. America

- Sweden
  - Blast furnace slag — 45 %
  - Steel slag — 100 %
  - RAP — 95 %
- Germany
  - Blast furnace slag — 100 %
  - Steel slag — 92 %
  - Coal bottom ash — 97 %
  - Coal fly ash — 88 %
  - RAP — 55 %

*RAP = recycled asphalt pavement

Source: (AMEC Case study, 2007)

- United States.
  - Blast furnace slag — 90 %
  - Coal bottom ash — 31 %
  - Coal fly ash — 27 %
  - RAP — 80 %



Recycled Slag, Paris, France

Highway 59 On a Good Day



Source: (http://www.texasfreeway.com/Houston/photos/59sw/59sw.shtml)

Highway 59 On a Bad Day



Source: (https://www.texasfreeway.com/houston/june-2001-flood-photos/)

A Smoggy day in Los Angeles



Los Angeles used to be the most polluted air basin in the United States.

Source: (California Air Resources Board)

A Clear day in Los Angeles



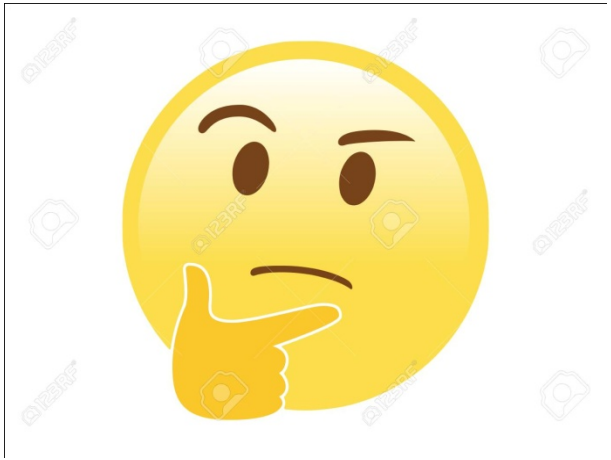
Source: (California Air Resources Board)

Conclusion

A few points to ponder!

- Negative externalities of non renewable resources
- Global warming & climate change
- Increasing awareness towards environmental sustainability
- Fossil fuels based economy vs Renewable energy centric economy
- Journey towards sustainability

Are we doing our share in the transportation sector?



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Thank You